

Pre-screening

Participant	012	
	Yes	No
Do you or have you had a history of migraine headaches?	☒	☐
Do you or have you had a history of claustrophobia?	☐	☒
Do you or have you had a history of frequent or severe motion sickness?	☐	☒
Do you or have you had a history of any health problems (e.g., seizures, diabetes, heart problems, vertigo) that affect your ability to drive?	☐	☒
If you are a female, are you or is there a possibility that you might be pregnant?	☐	☒

Demographics

- | | |
|--|--|
| 1. What is your race/ethnicity? | ☒ White / Caucasian |
| 2. What is your gender? | ☒ Female |
| 3. How old are you? | 39 |
| 4. What is the highest level of education you have completed? | ☒ 12th grade or less |
| 5. Which option best describes your current home? | ☒ I have no permanent residence. |
| 6. How do you pay for healthcare? | ☒ Private insurance |
| 8. Which best corresponds to your current work situation? | ☒ Working part time. |
| 9. What is your combined family income for the past 12 months (from all sources) | ☒ Less than \$5,000 |

Timeline Followback

Visit 1

Visit 1 Date

06-03-2025

Have you used opioids or any other substances during the assessment period?

☒ Yes

Visit 2

Visit 2 Date

06-18-2025

Have you used opioids or any other substances during the assessment period?

☒ Yes

Visit 3

Visit 3 Date

07-14-2025

Have you used opioids or any other substances during the assessment period?

☒ Yes

Visit 4

Visit 4 Date

07-30-2025

Have you used opioids or any other substances during the assessment period?

☒ Yes

Visit 5

Visit 5 Date

08-05-2025

Have you used opioids or any other substances during the assessment period?

☒ Yes

Phone

Participant # _____

012

System Usability Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently	☐	☐	☐	☐	☒
2. I found the system unnecessarily complex	☒	☐	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☐	☒
6. I thought there was too much inconsistency in this system.	☒	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☒
8. I found the system very cumbersome to use.	☒	☐	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☐	☒
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this app experience.	☒	☐	☐	☐	☐
2. The time I spent in this app just slipped away.	☐	☐	☐	☐	☒
3. I was absorbed in the app task.	☐	☐	☐	☐	☒
4. I felt frustrated while using the app.	☒	☐	☐	☐	☐

5. I found this app to be confusing.	☒	☐	☐	☐	☐
6. Using this app was taxing.	☒	☐	☐	☐	☐
7. The app is attractive.	☐	☐	☐	☐	☒
8. The app is aesthetically appealing.	☐	☐	☐	☐	☒
9. The app appealed to my senses.	☐	☐	☐	☐	☒
10. Using this app was worthwhile.	☐	☐	☐	☐	☒
11. The therapy experience was rewarding.	☐	☐	☐	☐	☒
12. I felt interested in the therapy tasks.	☐	☐	☐	☐	☒

iPAL Application and Intervention Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☐	☒
2. The videos and breathing exercises matched my internal or external triggers.	☒	☐	☐	☐	☐
3. The videos were videos easy to understand.	☐	☐	☐	☐	☒
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☐	☒

AR

Participant ID _____

012

- | | |
|---|---|
| 1. Which option describes your current home? | ☒ It is rented for money by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Private insurance |
| 3. Which best corresponds to your current work situation? | ☒ Working part time. |

iGROUP Presence Questionnaire

- | | |
|--|---|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Completely Disagree |
| 2. I felt that the virtual world surround me. | ☒ Completely Disagree |
| 3. I felt like I was perceiving pictures. | ☒ Completely Disagree |
| 4. I did not feel present in the virtual space. | ☒ Completely Agree |
| 5. I had a sense of acting in the virtual space, rather than operating something form the outside. | ☒ Completely Agree |
| 6. I felt present in the virtual space | ☒ Completely Disagree |
| 7. How aware were you of the real world surrounding you while navigating in the virtual world? | ☒ Very aware |
| 8. I was not aware of my real environment. | ☒ Completely Disagree |
| 9. I still paid attention to my real environment. | ☒ Completely Agree |
| 10. I was completely captivated by the virtual world. | ☒ Completely Disagree |
| 11 How real did the virtual world seem to you? | ☒ Very unreal |
| 12. How much did your experience in the virtual environment seem consistent with your real-world experience? | ☒ Completely Disagree |
| 13. How real did the virtual world seem to you? | ☒ Very unreal |
| 14. The virtual world seemed more realistic than the real world. | ☒ Completely Disagree |

System Usability Scale AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☒	☐	☐	☐	☐
2. I found the system unnecessarily complex.	☒	☐	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☐	☒
6. I thought there was too much inconsistency in this system.	☐	☐	☐	☐	☒
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☒
8. I found the system very cumbersome to use.	☐	☐	☐	☐	☒
9. I felt very confident using the system.	☐	☐	☐	☐	☒
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this augmented reality experience	☒	☐	☐	☐	☐
2. The time I spent in augmented reality just slipped away.	☒	☐	☐	☐	☐
3. I was absorbed in the therapy task.	☒	☐	☐	☐	☐
4. I felt frustrated while using the headset.	☐	☐	☐	☒	☐
5. I found this system to be confusing.	☒	☐	☐	☐	☐
6. Using this system was taxing.	☐	☐	☐	☐	☒
7. The virtual environment is attractive	☒	☐	☐	☐	☐

8. The augmented reality is aesthetically appealing.	☒	☐	☐	☐	☐
9. The augmented environment appealed to my senses.	☒	☐	☐	☐	☐
10. Using this system was worthwhile.	☐	☐	☐	☒	☐
11. The therapy experience was rewarding.	☐	☐	☐	☒	☐
12. I felt interested in the therapy tasks.	☒	☐	☐	☐	☐

iPAL Application and Intervention Evaluation AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☒	☐	☐	☐	☐
2. The videos and breathing exercises matched my internal or external triggers.	☒	☐	☐	☐	☐
3. The videos were clear and easy to understand.	☐	☒	☐	☐	☐
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☐	☒

VR

Participant ID _____

12

- | | |
|---|---|
| 1. Which option describes your current home? | ☒ I have no permanent residence |
| 2. How do you pay for healthcare? | ☒ Private insurance |
| 3. Which best corresponds to your current work situation? | ☒ Working part time. |

iGroup Presence Scale

- | | |
|--|---|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Somewhat Agree |
| 2. I felt that the virtual world surrounded me. | ☒ Somewhat Agree |
| 3. I felt like I was perceiving pictures. | ☒ Somewhat Agree |
| 4. I did not feel present in the virtual space. | ☒ Somewhat Disagree |
| 5. I had a sense of acting in the virtual space, rather than operating something from the outside. | ☒ Somewhat Agree |
| 6. I felt present in the virtual space. | ☒ Somewhat Agree |
| 7. How aware were you of the real world surrounding you while navigating in the virtual world? | ☒ Somewhat aware |
| 8. I was not aware of my real environment. | ☒ Completely Disagree |
| 9. I still paid attention to my real environment. | ☒ Completely Disagree |
| 10. I was completely captivated by the virtual world. | ☒ Somewhat Agree |
| 11. How real did the virtual world seem to you? | ☒ Somewhat real |
| 12. How much did your experience in the virtual environment seem consistent with your real-world experience? | ☒ Somewhat Disagree |
| 13. How real did the virtual world seem to you? | ☒ Somewhat unreal |
| 14. The virtual world seemed more realistic than the real world. | ☒ Somewhat Disagree |

System Usability Scale VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☐	☐	☐	☐	☒
2. I found the system unnecessarily complex.	☐	☒	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☐	☒
6. I thought there was too much inconsistency in this system.	☒	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☒	☐
8. I found the system very cumbersome to use.	☐	☐	☐	☒	☐
9. I felt very confident using the system.	☐	☐	☐	☐	☒
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this virtual experience	☐	☐	☐	☒	☐
2. The time I spent in the virtual world just slipped away.	☐	☐	☐	☒	☐
3. I was absorbed in the therapy task.	☐	☐	☒	☐	☐
4. I felt frustrated while using the headset.	☒	☐	☐	☐	☐
5. I found this system to be confusing.	☒	☐	☐	☐	☐
6. Using this system was taxing.	☒	☐	☐	☐	☐
7. The virtual environment is attractive	☐	☐	☐	☒	☐

8. The virtual environment is aesthetically appealing.	☐	☐	☐	☒	☐
9. The virtual environment appealed to my senses.	☐	☐	☐	☒	☐
10. Using this system was worthwhile.	☐	☐	☐	☒	☐
11. The therapy experience was rewarding.	☐	☐	☒	☐	☐
12. I felt interested in the therapy tasks.	☐	☐	☒	☐	☐

iPAL Application and Intervention Evaluation VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☒	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☐	☒
3. The videos were clear and easy to understand.	☐	☐	☐	☐	☒
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☒	☐

MR

Participant ID _____

012

- | | |
|---|---|
| 1. Which option describes your current home? | ☒ It is rented for money by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Private insurance |
| 3. Which best corresponds to your current work situation? | ☒ Working part time. |

iGROUP Presence Questionnaire MR

- | | |
|--|---|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Somewhat disagree |
| 2. I felt that the virtual world surrounded me. | ☒ Somewhat Agree |
| 3. I felt like I was perceiving pictures. | ☒ Somewhat Agree |
| 4. I did not feel present in the virtual space. | ☒ Somewhat Agree |
| 5. I had a sense of acting in the virtual space, rather than operating something from the outside. | ☒ Somewhat Agree |
| 6. I felt present in the virtual space. | ☒ Somewhat disagree |
| 7. How aware were you of the real world surround you while navigating in the virtual world? | ☒ Very aware |
| 8. I was not aware of my real environment. | ☒ Completely disagree |
| 9. I still paid attention to my real environment. | ☒ Completely Agree |
| 10. I was completely captivated by the virtual world. | ☒ Somewhat disagree |
| 11. How real did the virtual world seem to you? | ☒ Somewhat unreal |
| 12. How much your experience in the virtual environment seem consistent with your real-world experience? | ☒ Somewhat disagree |
| 13. How real did the virtual world seem to you? | ☒ Somewhat unreal |
| 14. The virtual world seemed more realistic than the real world. | ☒ Completely disagree |

System Usability Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☒	☐	☐	☐	☐
2. I found the system unnecessarily complex.	☐	☐	☐	☒	☐
3. I thought the system was easy to use.	☐	☐	☐	☒	☐
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☒	☐	☐	☐
6. I thought there was too much inconsistency in this system.	☐	☒	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☒	☐
8. I found the system very cumbersome to use.	☐	☐	☐	☒	☐
9. I felt very confident using the system.	☐	☐	☐	☒	☐
10. I needed to learn a lot of things before I could get going with this system.	☐	☒	☐	☐	☐

User Engagement Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this mixed-reality experience.	☒	☐	☐	☐	☐
2. The time I spent in mixed reality just slipped away.	☒	☐	☐	☐	☐
3. I was absorbed in the therapy task.	☒	☐	☐	☐	☐
4. I felt frustrated while using the headset.	☐	☐	☐	☒	☐
5. I found this system to be confusing.	☐	☒	☐	☐	☐
6. Using this system was taxing.	☐	☐	☐	☒	☐
7. The virtual environment is attractive	☐	☒	☐	☐	☐

8. The mixed-reality is aesthetically appealing.	☐	☒	☐	☐	☐
9. The blended environment appealed to my senses.	☐	☒	☐	☐	☐
10. Using this system was worthwhile.	☐	☒	☐	☐	☐
11. The therapy experience was rewarding.	☐	☒	☐	☐	☐
12. I felt interested in the therapy tasks.	☐	☒	☐	☐	☐

iPAL Application and Intervention Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☒	☐	☐	☐	☐
2. The videos and breathing exercises matched my internal or external triggers.	☒	☐	☐	☐	☐
3. The videos were clear and easy to understand.	☐	☐	☐	☐	☒
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☐	☒

Exit Interview

1. Which option describes your current home?	☒ It is rented for money by you or someone in household
2. How do you pay for healthcare?	☒ Private insurance
3. Which best corresponds to your current work situation?	☒ Working part time.
1. You experienced Cognitive Behavioral Therapy and Heart Rate Variance interventions in virtual, augmented, and mixed realities. How would you describe your experiences with each of those headsets?	enjoyed VR the most because it actually felt like the participant was going to a different place. disliked both AR and VR. felt like it interfered with vision and caused difficulty focusing on the task because of being so aware of the surroundings.
2. Do you think these types of technologies have a place in OUD treatment and management? If YES, how? If NO, why not?	yes - only VR. could envision VR being used to calm an individual down during crisis, especially with the breathing exercise. if an individual was truly focusing, the CBT videos could be helpful as well.
3. How would you describe the iPAL application, in terms of a cravings management tool?	feels like the app could be useful because it took the participant's mind off of the cravings, believes the questions asked were appropriate but feels like more questions could have been asked and could have delved deeper.
4. How would you describe the interventions (CBT, HRV)? Were the interventions appropriate for the intensity level and triggers you reported?	CBT - feels like it would be helpful to an individual that is new to recovery. feels like the intervention was not super beneficial for them because they have been in recovery for so long and have heard the same things in the videos before. HRV - felt like the intervention was too long. could have been helpful to have a break in between and feels like it may be helpful to incorporate different breathing exercises into the activity, such as the 4x4 breathing exercise.
5. Do you think the iPAL app, or similar digital interventions delivered via phone should be integrated into OUD treatment plans? If YES, how and in what capacity or role? if NO, why not?	yes - feels like it would be beneficial to take it home and utilize it daily or whenever they are feeling increased tension or anxiety. feels like this should be something that is utilized as needed depending on the individual patient and their stability
6. If iPAL or a similar app was available, would you download the app and incorporate it into your OUD treatment and management plan?	yes - could envision self utilizing the app a couple of times per week